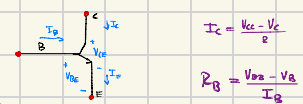


BJT Power Problems



$P_{BJT} = I_B V_{BE} + I_C V_{CE}$ (on BJT)

$P_{load} = I^2 \cdot R$, $P_{load} = \frac{V^2}{R}$ (Load)

Total Harmonic Distortion

$$= 100\% \left[\frac{(A_1)^2 + (A_2)^2 + \dots + (A_n)^2}{(A_1)^2} \right]$$

Filter Specifications

$$\frac{A_{y, min} - A_s}{\log_{10}(A_s) - \log_{10}(A_p)} \leq 20n$$

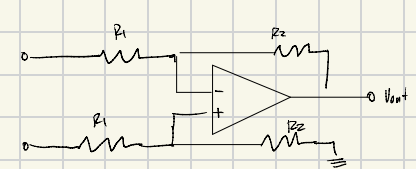
$\therefore n = \frac{20}{14 \dots 2}$

(n, as whole number $\rightarrow 14 \dots 2$)

Common mode gain (A_{cm})

$$A_{cm} = \frac{V_{ocm}}{V_{cm}} < 1$$

Example: Op-amp circuit (CMRR)



$V_{out} = V_{odm} = A_{dm}(V_2 - V_1) = \frac{R_2}{R_1}(V_2 - V_1)$

Instantaneous power and average power

(y) $dBm = 10 \log_{10} \left(\frac{x \text{ mW}}{1 \text{ mW}} \right)$

(x) $mW = (1 \text{ mW}) (10^{\frac{y}{10}})$

How many Watts is across a 50-ohm resistor dissipating -10 dBm or Power?

$P_{mW} = 10^{\frac{-10}{10}} \rightarrow 10^{-1} \text{ mW} = 10^{-1} \cdot 10^{-3} \text{ W}$
 $= 10^{-4} \text{ W} = 0.0001 \text{ W}$

$P = \frac{V^2}{R} \rightarrow V = \sqrt{P \cdot R} = \sqrt{(0.0001) \cdot 50} = 0.0707 \text{ V} \rightarrow 70.7 \text{ mV}$

Common mode Voltage

$$V_{cm} = \frac{V_1 + V_2}{2}$$

Differential mode Voltage

$$V_{dm} = \frac{V_1 - V_2}{2}$$

Given V_{cm} and V_{dm} we may represent V_1 and V_2 as

$V_1 = V_{cm} + V_{dm}$
 $V_2 = V_{cm} - V_{dm}$

Common Mode Output Voltage

If $V_1 = V_2 = V_{cm}$, then $V_o = 0$

$$V_o = A_{oc}(V_1 - V_2) + V_{ocm}$$

CMRR Example

notice that $A_{om} = \frac{R_2}{R_1} = \frac{10k\Omega}{1k\Omega} = 10$, and given open loop CMRR of 40 dB we can solve for A_{cm} as follows,
 $CMRR = 1025 = 20 \log_{10} \left(\frac{A_{om}}{A_{cm}} \right)$
Then,
 $31623 = \frac{A_{om}}{A_{cm}}$ and, $A_{cm} = \frac{A_{om}}{31623} = \frac{10}{31623}$

CMRR Example (cont.)

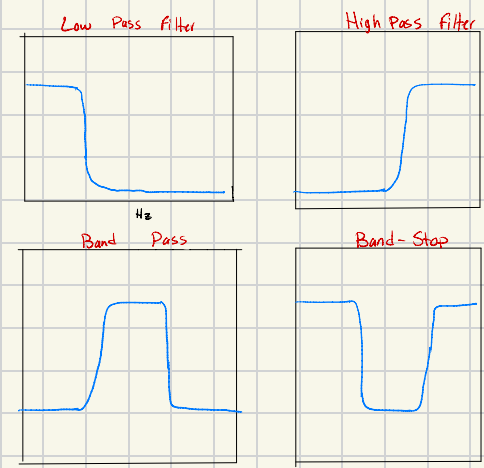
Finally,
 $V_{ocm} = A_{cm} V_{cm}$
 $= \left(\frac{10}{31623} \right) \cdot 2 \text{ mV}$
 $= 0.63 \mu\text{V} @ 50 \text{ kHz}$

Filter Families

- There are many types of filter families, such as,
 - Butterworth
 - Chebyshev
 - Bessel
 - Cauer / Elliptic

Filter Parameters

- The differences in filter families is how each one chooses to optimize some filter parameters.
- Out of a given number of parameters, a 'family' will optimize some desired parameter/s, at the expense of some other parameters.
- Multiple filter parameters exist, but we will focus on only three:
 - Gain flatness in the passband.
 - Steepness of the roll-off in the transition region.
 - Linear phase in the passband.



Best Practices! – Silkscreen

- Finest pen width: **6 mils**
- Smallest character height: **50 mils**
- Recommended height-to-pen-width ratio: **6-10:1**
- Highly recommended to include:
 - Unique board ID (identifier/name)
 - Identification (Division/Team/Your)
 - Rev. identifier or Date code.

Types of PCBs

of layers = # of metal layers

- Single-Sided PCB
- Double-Sided PCB
- Multilayer PCB
- Layers: 1-2-4-6-8-etc.

[Prepreg are sheets of fiber glass, pre-impregnated with epoxy resin.]

Filter Family Comparison

- Three commonly encountered filter families are,
 - Butterworth
 - Chebyshev
 - Bessel

(in passband)	Chebyshev	Butterworth	Bessel
Gain Flatness	Has ripple	Monotonically Decreases	Monotonically Decreases
Sharpness of corner at cutoff	Sharpest	Medium	Least Sharp
Group Delay Flatness	Worst	Medium	Best
# of stages for given amplitude	Least	Medium	Most

PCB Parasitics – A Necessary Task

As the signal rise time of signals decrease and signals become more sensitive to noise, the interconnects may no longer be transparent.

Question: What is the impact of the interconnects?

- Principle vs. Practice
- Performance/Design Margins
- To determine if interconnects are transparent, you will need to calculate the equivalent circuit element using their geometry and material property.
 - Signal Integrity relates to how the interconnects introduce parasitics and analyzing their impacts.
 - See PCB Traces and Signals Propagation quiz for more information.

PCB Terminology

- Layers** – Refers to the number of conducting layers (typically copper) within the board (but realistically there are several layers of different material).
 - Substrate** – A fiberglass layer (FR4) that provides board rigidity and thickness.
 - Copper** – The conductive layer which is shaped/routed to provide 'traces'.
 - Solder mask** – A lacquer-like layer of protection that prevents oxidation and short circuits.
 - Silkscreen** – A layer of ink used to identify the PCB components, marks, logos, symbols, etc.
- Solder bridge** – An unintended connection between two conductor made by a small blob of solder.

PCB Terminology (cont.)

- Pads** – The area of the PCB that has exposed metal where the components connect to (typically by soldering the component).
- Via/Through hole** – A plated hole wall that goes through part of the PCB that provides a path that can connect the electrical signal from one layer to another (typically going from the top all the way to the bottom. [See also 'blind via' and 'buried via'.])
- Annular ring** – A ring of copper that encircles the via.
- Copper Traces** – segments of the copper layer that have been 'traced' out to make a continuous path.
- Footprint** – The arrangement of pads/through-holes used to describe how/where the component will connect to the PCB.

Thermal Analysis Management

Consider the three metrics:

1. Dissipated power P_D [Watts] I
2. Temperature gradient ΔT [Degrees Celsius] ΔV
3. Thermal resistance θ [Degrees Celsius/Watts] R

• These are related with one another, analogous to how current, voltage and resistance are related with one another.

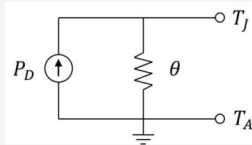
• We use the following expression to find junction/device temperature.

$$T_j = T_A + \Delta T = T_A + P_D \theta$$

Thermal Analysis Management (cont.)

- Example: Given that the ambient temperature is $T_A = 25^\circ\text{C}$, dissipated power is $P_D = 10\text{W}$ and thermal resistance is $\theta = 20^\circ\text{C/W}$ what is the device temperature.

$$\begin{aligned} T_j &= T_A + \Delta T \\ &= T_A + P_D \theta \\ &= 25 + (10)(20) \\ &= 225^\circ\text{C} \end{aligned}$$



Power Efficiency BJT Circuit

$$\% = \frac{\text{Power Load}}{P_{BJT} + P_{Load}} \times 100\%$$

BJT Power (cont.)

$$I_E = I_B + I_C$$

$$I_E = \alpha I_C$$

$$\alpha = \frac{\beta}{\beta + 1}$$

Pre-Lab Exercise 06
Compute voltages V_a , V_b , V_{cm} , V_{dm} where V_{cm} and V_{dm} is common mode voltage and differential mode voltage between V_a and V_b , respectively.
 $V_{cm} = \frac{V_1 + V_2}{2}$

1) $i_1 = 0.7\text{mA}$, $i_2 = 0.7\text{mA}$, $V = 2.8\text{V}$, $V_a = 1.4\text{V} = V_b$, $V_{cm} = 1.4\text{V}$, $V_{dm} = 0\text{V}$

2) $i_1 = 0.7015\text{mA}$, $i_2 = 0.6973\text{mA}$, $V_a = 1.405\text{V}$, $V_b = 1.415\text{V}$, $V_{cm} = 1.41\text{V}$, $V_{dm} = -0.005\text{V}$

3) $i_1 = 1.4\text{mA}$, $V_a = V_b = 4.2 - 1.4 = 2.8\text{V}$, $V_{cm} = 2.8\text{V}$, $V_{dm} = 0\text{V}$, $V_{tot} = 4.2\text{V}$

4) $i_1 = 1.0525\text{mA}$, $i_2 = 1.0447\text{mA}$, $V_a = 4.21 - 2.105 = 2.105\text{V}$, $V_b = 4.21 - 2.0893 = 2.12\text{V}$, $V_{cm} = 2.1125\text{V}$, $V_{dm} = -0.0075\text{V}$